GABRIEL GUARD

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EDUCATION

The University of Texas at Austin ***May 2030***
B.S. in Mechanical Engineering | Intended Minor in Robotics

RELEVANT COURSEWORK

AP Physics C Mechanics, AP Physics C E&M, AP Physics 1, AP Calculus BC, AP Calculus AB, AP Research, AP Statistics, AP Chemistry, AP Computer Sci. A, Intro to Engineering, Aerospace Engineering, Comp Sci Principles, Practicum in STEM

EXPERIENCE

Competitive Robotics (FIRST FRC), Team 6800 | 1,020 hours **Aug 2023 - May 2026**
Mechanical Design Lead

- Designed 4 complete robot subsystems using top-down, parametric CAD in Onshape, creating detailed technical drawings with tolerances and optimizing designs for manufacturability, assembly, and performance while managing development through Jira.
- Solely led the manufacturing and assembly of a robot subsystem, managing part inventory, orders, and BOMs; verified CAM toolpaths, oversaw machining, inspected finished parts, and assembled the final subsystem.
- Programmed CAM toolpaths in VCarve and Fusion 360 for CNC router and mill manufacturing.

Manufacturing

- Operated CNC router, manual lathe, 3D printers and hand tools to produce custom robot parts.
- Gaining proficiency in manual and CNC milling to expand machining capabilities.

Competition Driver

- Selected as primary competition driver, leading match strategy, coordinating alliance partners, and communicating under pressure to adapt strategy and execute coordinated match plans on the world stage

Achievements
(Team) FIRST World Championship Division Winner, FIRST World Championship Division Finalist, FIRST In Texas District Championship Winner, FIRST In Texas District Championship Finalist, 4x District Event Winner, 6x Industrial Design Award Winner, 2x Excellence in Engineering, 1x Engineering Inspiration, (Individual) Team's Choice Award

Competitive Robotics (FIRST FTC), Team 11503 **Aug 2022 - May 2023**

- Designed lift and manipulator mechanisms; 3D printed and assembled subsystem
- Selected to drive robot at competitions and coordinate match strategy

PROJECTS

Valor CAD Challenge | 30 hours

- Organized an FTC design challenge with 117 participants, designing and modeling a mock FTC field, generating a game manual, and judging 28 submissions for practicality and feasibility.
- Produced and livestreamed a sponsored awards ceremony, reaching 750 viewers while coordinating award presentations and event logistics.

Umbra Design LLC | 55 hours

- Founded an LLC developing and selling proprietary translucent car sun visors, taking the product from concept through manufacturing and assembly using 3D printing and laser cutting.
- Pursuing a utility patent while optimizing the design for manufacturability and producing functional units through iterative prototyping.

Valor STEM Reach | 8 hours

- Presented to middle school STEM classes to increase robotics exposure, impacting 200+ students
- Hosted 50 elementary school students for a robotics season kickoff

SKILLS

Certifications: Onshape Associate, Autodesk Inventor, Microsoft Office Associate (Excel, Word, PowerPoint), OSHA 10
Proficiencies: (CAD) Onshape, Inventor, Fusion360, (Management) Jira, Confluence, (Manufacturing) Router, Laser Cutter, 3D Printer, (Graphic Design) Figma, (Spanish) B1 Threshold
Pursuing: SolidWorks, Onshape Professional, MATLAB, CNC Mill